Systems Analysis & Design

Semester 2026-I

Workshop No. 2

Systems Analysis - Local business support recovery

Group Seven

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Executive Summary

This project presents the design of a support system for the recovery of small local businesses in the Titan Plaza shopping mall in Bogotá, based on the analysis conducted in Workshop #1. That analysis identified key issues such as the low conversion of visitor traffic into sales, the limited visibility of small businesses, and the lack of tools to properly manage inventory and adapt to customer behavior.

Although the shopping mall experiences a high flow of visitors, especially in entertainment and dining areas, this traffic does not effectively translate into benefits for small businesses. This is largely due to the lack of information integration and the disadvantage these businesses face compared to large retail chains, which have more advanced technological capabilities and more consolidated market strategies.

As a solution, the design of an analytical platform is proposed to help small business owners better understand customer behavior, optimize their inventory decisions, and improve their visibility within the shopping mall. This system aims to strengthen data-driven decision-making, enabling local businesses to increase their profitability and popularity, thereby reducing the gap between them and large retail chains.

Requirements Analysis

Based on the findings from Workshop #1, several key requirements were identified to address the challenges faced by small businesses in Titan Plaza. The system must bridge the gap between customer demand and business supply, improve their visibility, and enable more informed decision-making through the use of data.

Functional Requirements

- Customer Traffic Information

  The system must inform users through surveys to other users about customer traffic, considering factors such as weather, day of the week, and time. A significant increase in traffic is observed depending on the day of the week and time.

- Inventory Alerts

  The system must notify owners when a product has low turnover or is running out. Although no store reported low inventory frequency, it is important that no store ever runs out of stock of any product.

- Automatic Promotions

  The system must send promotions. 41.18% of customers spend less now than before the pandemic. Promotions encourage them to buy and increase the visibility of many businesses that are not so visible in the shopping mall.

- Dashboard for Owners

  The system must display key metrics: sales, traffic, inventory turnover in real time. Owners need to make informed decisions.

- Product Recommendations

  The system must suggest products to customers based on previous purchases and trends. Improve conversion rate.

Non-Functional Requirements

- Security

  Customer and sales data must be encrypted and protected.

- Usability

  The interface must be intuitive for store owners; not everyone is a technology expert.

- Availability

  The system must have 99.5% uptime during business hours (8:00 a.m. to 10:00 p.m.).

- Scalability

  It must support growth to additional shopping malls without requiring a major redesign.

- Portability

  The system must work on mobile devices (Android/iOS) and web.

Constraints

- Data protection

  Must comply with Colombian personal data protection laws (Law 1581 of 2012).

- Owner distrust

  Not all store owners are willing to start using a system and share private information about their businesses.

- Connectivity

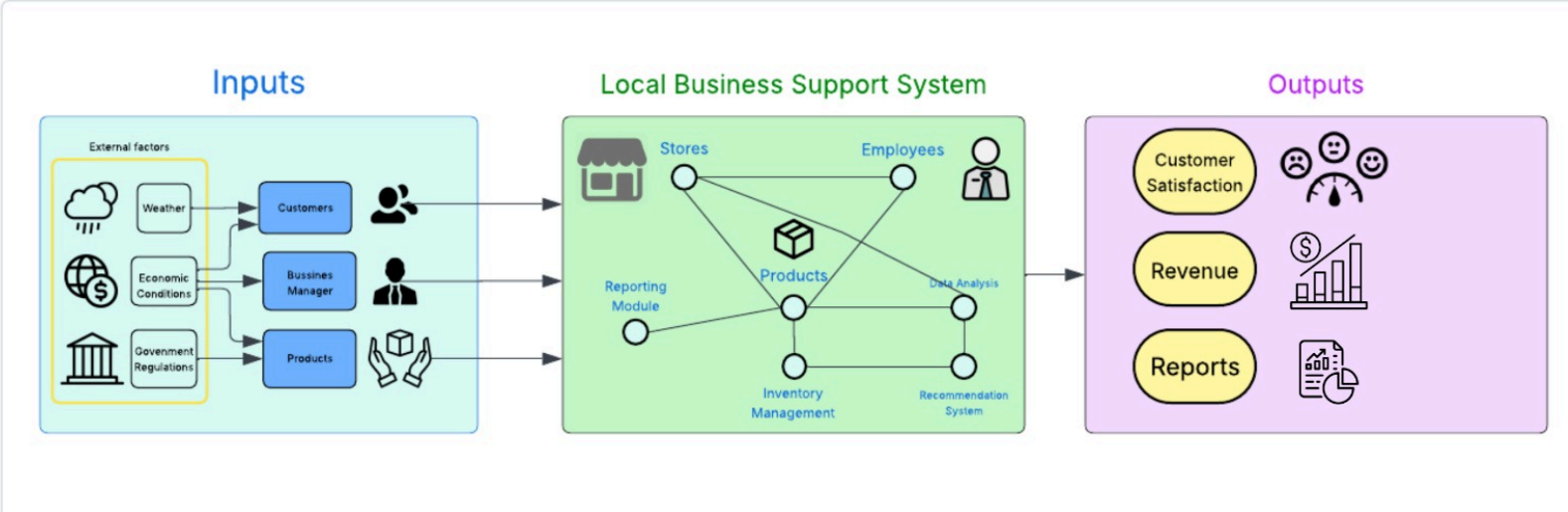
  Some areas of the mall have variable internet signal; must work partially offline.

- Existing systems

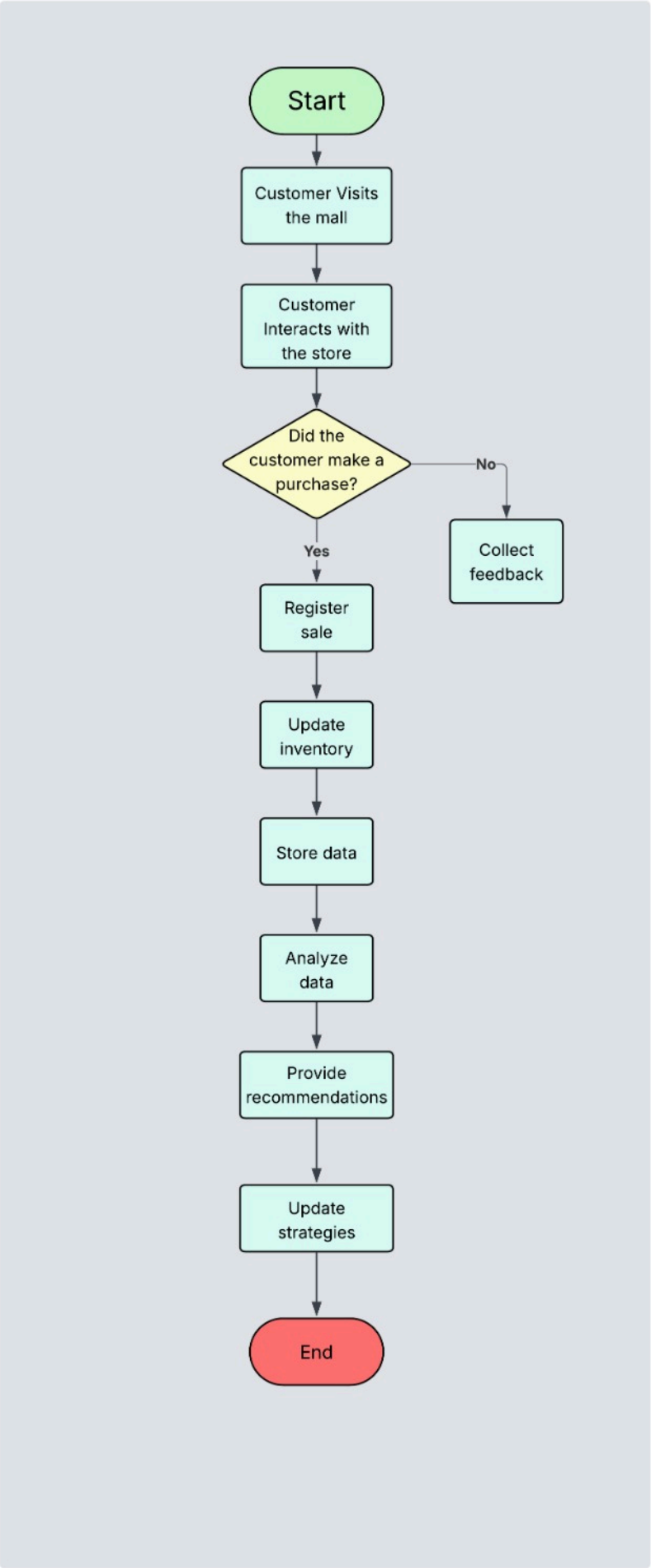
  Many owners already use dedicated inventory rotation systems; the solution must be able to integrate with them or coexist without causing conflicts.

System Architecture

System Architecture Diagram



PROCESS FLOW DIAGRAM



Architectural Overview

The system is based on a modular architecture centered on the **Local Business Support System**, which integrates data collection, processing, and visualization.

It consists of three main layers:

- **Data Input:** customer, business manager and products
- **Processing:** Local business support system
- **Output:** Revenues and reports for decision-making

This design is simple, scalable, and suitable for small business environments.

System Components

The main components of the system are:

- **Customers:** generate data through interactions and purchases
- **Business Owner / Manager:** makes decisions based on system outputs
- **Store Operations:** core business activities
- **Employees:** handle customer interactions and sales
- **Inventory Management:** controls stock levels
- **Sales Module:** records transactions
- **Data Analysis Module:** processes and analyzes data
- **Recommendation System:** suggests actions
- **Reporting Module:** presents information

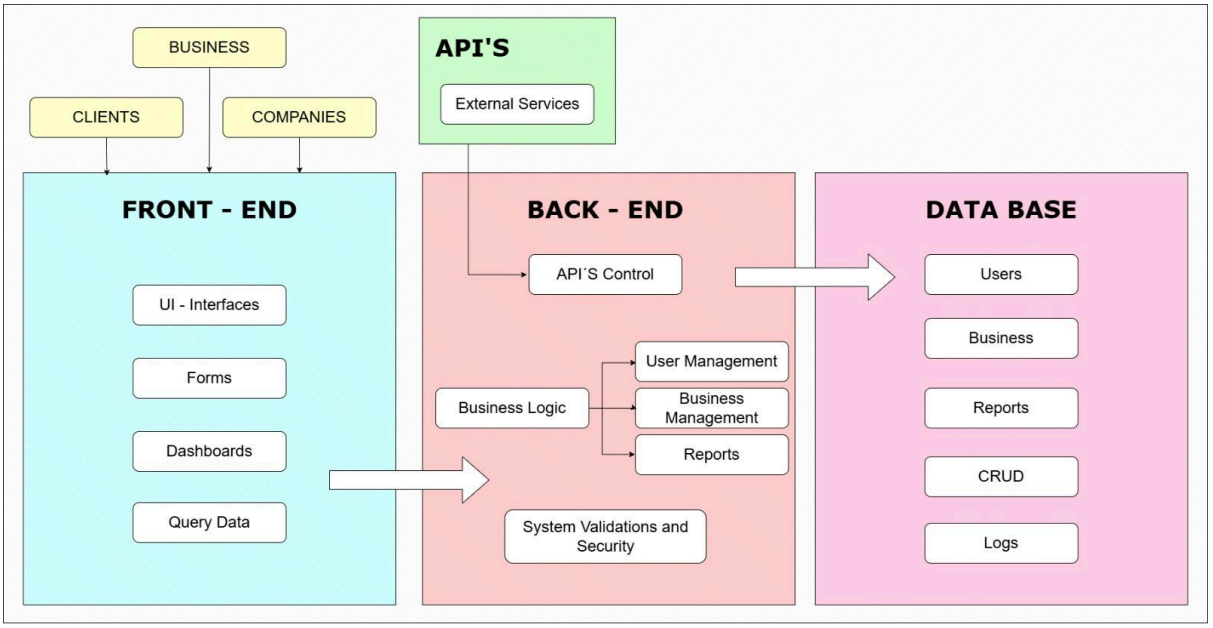
Data Flow Design

The process starts when customers interact with the store. If a purchase is made, the sale is recorded and inventory is updated. If not, feedback can be collected.

All data is stored and then analyzed to identify patterns such as demand trends and customer preferences. The system generates insights and recommendations, which are used by the business owner to adjust strategies.

This creates a continuous improvement cycle based on data.

Interfaces and Integration



Complexity and Risk Management

Identified Risks

About 41% of customers are spending less than before the pandemic, which makes it harder to keep them engaged with local businesses. Foot traffic varies a lot: Sundays bring in more than double the visitors compared to Saturdays, and weather also plays a big role, making it tough for store owners to manage inventory properly. Nearly 30% of visitors come to the mall for reasons other than shopping, so reaching them with promotions isn't easy. There's also the risk of running out of popular products if inventory alerts don't work well. On top of that, many store owners are reluctant to share their business data, and some areas of the mall have poor internet connection, which could affect how often people use the platform.

Mitigation Strategies

We'll send more personalized promotions, especially during peak traffic days. The dashboard will show traffic predictions so owners can prepare ahead and stock up on what sells most. Inventory alerts will notify them before products run out. We'll also provide hands-on training to help owners feel more comfortable with the system, and the interface will be kept simple. For areas with spotty internet, the platform will work offline and sync later. And for those already using other systems, we'll make sure everything can integrate smoothly.

Monitoring and Control Mechanisms

We'll track weekly how many owners are using the dashboard and whether they're satisfied with it. Promotion effectiveness will be measured by looking at sales increases and customer response. Predictions will be compared to actual foot traffic to fine-tune the model. We'll keep an eye on inventory alerts to see if they actually prevent stockouts. Connection errors

will be logged to improve offline performance, and regular audits will ensure customer and business data stays protected.

Technical Implementation Strategy

Technologies and Tools

The platform is structured across three main layers, as illustrated in the System Architecture Diagram. The front-end layer handles all user-facing components, including the business dashboard, data query interfaces, and reporting forms that store owners interact with directly. The back-end layer manages the core business logic, covering user management, business management, report generation, system validations, and security controls, as well as the integration with external services through API connections. Finally, the database layer stores all persistent information including user records, business data, reports, operation logs, and CRUD interactions. This three-layer separation ensures modularity and maintainability, allowing each component to be updated or scaled independently as the platform grows beyond Titan Plaza.

Development Methodology

The system will be developed following an iterative and incremental approach, where each cycle delivers a functional subset of the platform that can be tested and validated by store owners. This is particularly important given that a significant portion of business operators have limited familiarity with technology, meaning the interface and features must be progressively refined based on real user feedback rather than delivered all at once. Each iteration will prioritize the most critical functionalities first — such as the dashboard and inventory alerts — before advancing toward more complex features like demand forecasting and recommendation systems. This approach also reduces the risk of building components that do not align with actual operational needs, ensuring that the final solution remains practical, usable, and directly responsive to the challenges identified during the systems analysis of Titan Plaza.

Deployment Strategy

The system will be deployed in a controlled and gradual manner, beginning with a pilot phase involving a small group of volunteer stores within Titan Plaza. This initial rollout allows the team to validate the platform under real operational conditions, identify integration issues with existing business workflows, and build trust among store owners before broader adoption. During this phase, participating businesses will receive direct support to ensure a smooth onboarding experience. Once the pilot results are validated and necessary adjustments are incorporated, the platform will be made available to all small businesses within the mall. Given that some areas of Titan Plaza present variable internet connectivity, the deployment process will include offline synchronization mechanisms that allow the system to continue operating under limited network conditions and sync data once connection is restored. As the platform matures and demonstrates measurable value, the architecture is designed to support expansion to additional shopping malls with minimal reconfiguration, treating each new location as an independent operational unit within the same infrastructure.

Implementation Phases

The development and deployment of the platform will be structured across four sequential phases. The first phase focuses on the foundational setup of the system, including the definition of the database schema, the core business logic, and the basic structure of the owner dashboard with read-only metrics. The second phase introduces the active modules of the platform, particularly the sales recording system, the inventory management module, and the automated alert mechanism that notifies owners when stock levels are critical. The third phase incorporates the analytical layer, integrating the demand forecasting model and the product recommendation engine built upon the traffic and behavioral patterns collected during Workshop #1. The fourth and final phase covers the pilot deployment with selected stores, the collection of user feedback, performance tuning, and the full rollout to all small businesses within Titan Plaza, accompanied by training sessions to ensure proper adoption across the different levels of technological familiarity among store owners.

Performance and Optimization

Performance Metrics

The system's performance will be evaluated through a set of measurable indicators directly tied to the challenges identified in Workshop #1. Sales conversion rate will be tracked to determine whether the platform effectively translates visitor traffic into purchases for small businesses. Inventory stockout frequency will be monitored to assess whether the alert system successfully prevents product unavailability. Dashboard response time will be measured to ensure that store owners receive real-time data without significant delays. Additionally, promotion engagement rate will be tracked to evaluate whether the automated promotional features generate a measurable increase in customer interaction and spending, particularly relevant given that 41.18% of surveyed customers reported spending less than before the pandemic.

Optimization Strategies

The system will implement caching mechanisms for frequently accessed data such as current traffic estimates and top-selling products, reducing server load and improving dashboard responsiveness. The demand forecasting model will be continuously retrained as new sales and traffic data is collected, progressively improving the accuracy of its predictions over time. Promotions will be dynamically adjusted based on observed customer behavior patterns, ensuring that incentives are relevant and timely rather than static. For areas of the mall with poor connectivity, data synchronization will be optimized to prioritize critical updates — such as inventory changes and sales records — over less time-sensitive information.

Scalability Considerations

The platform is designed to scale horizontally as the number of participating stores grows, both within Titan Plaza and across future mall locations. Each shopping mall will operate as an independent unit within the shared infrastructure, ensuring that the growth of one location does not affect the performance of others. The analytical component will be deployed as a separate service, allowing it to be scaled independently from the rest of the system during periods of high data processing demand, such as weekends when foot traffic at Titan Plaza reaches its peak.

Conclusion and Future Work

The design presented in this document represents a direct response to the structural challenges identified during the systems analysis of Titan Plaza in Workshop #1. The core finding of that analysis — that small local businesses suffer from an informational disconnect that prevents them from competing effectively with large retail chains — has been addressed through the design of an analytical platform that centralizes data, automates critical alerts, and enables data-driven decision-making for store owners who previously relied solely on intuition and informal information flows.

The proposed architecture balances technical robustness with practical feasibility, taking into account the real constraints of the environment: variable internet connectivity, limited technological familiarity among store owners, existing inventory systems that must be respected, and the legal obligations imposed by Colombian personal data protection regulations. Rather than proposing an overly complex solution, the design prioritizes usability and gradual adoption as the primary drivers of long-term success.

Looking ahead, several opportunities exist for the continued evolution of the platform. As more sales and traffic data is accumulated over time, the accuracy of the demand forecasting and recommendation models will improve significantly, generating increasingly precise insights for store owners. Future development cycles could also explore the integration of loyalty programs and direct customer-facing features, transforming the platform from a purely internal business tool into a broader ecosystem that connects customers and merchants within the mall. Finally, the multi-mall expansion model opens the possibility of generating cross-location benchmarks, allowing store owners to compare their performance against similar businesses in other shopping centers and identify further optimization opportunities.

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